Table1: Configuration parameters

Parameter	Value for Left Box	Value for Right Box
Name	*TowardsVasantika*	*TowardsSankalp*
Remote host/IP	*192.168.51.1*	*192.168.20.254*
Remote subnet:	*192.168.51.0/255.255.255.0*	*192.168.20.0/255.255.255.0*
Host IP address	*Office1 RED IP Address*	*Office 2 RED IP Address*
Local subnet	*192.168.20.0/255.255.255.0*	*192.168.51.0/255.255.255.0*
Use a Pre-Shared Key (PSK)	*Thisisatestpresharedkey*	*Thisisatestpresharedkey*

Figure 1: Configuring site-to-site VPNs

Figure 2: Left VPN Tunnel Open

Figure 3: Right VPN Tunnel Open

name, and click the *Generate Root/Host Certificates* button. Fill in the required details. Download the *Host* and *Root* certificates of both the boxes.

On the left IPCop, go to the *VPNs* > *VPNs* menu and add a new Net-to-Net VPN connection. Set the *Remote Host IP* as the RED IP of the Right IPCop, and *Remote Subnet* as 192.168.51.0/255.255.255.0. Upload the Right IPCop's host certificate. Save this set-up. Upload the Right IPCop's root certificate via the *VPNs* > *VPNs* menu, giving the correct CA name used while creating the certificate.

Repeat the same process on the Right IPCop box, using the corresponding Left IPCop host and root certificates.

That is all! If the configuration is correct, the tunnel will immediately open. You may verify this from *Connection status* and *Control* under the *VPNs* > *VPNs* menu.

IPCop offers an excellent cost-effective solution to establish site-to-site VPNs. Once established, the connectivity is very stable, and requires attention only on rare occasions.

Important notes and troubleshooting guidelines:

- Make sure that different IP subnets are used in the offices.
- The default value in the *'Public IP or FQDN for RED interface'* VPN set-up field could have the service provider's name, such as *IPADDRESS.static-xxx.serviceprovider.net.in.* Make sure you remove non-IP-address text.
- IPCop RED with private IP addresses (such as 192.168.x.x or 10.x.x.x) makes VPN configuration really difficult. This happens when IPCop is installed inside a router configured in route mode. Here, reconfigure the router in bridge mode, so that IPCop RED will get a public IP.
- Refresh the browser a few times if tunnel 'open' status is not displayed.
- Ensure correct uploading of root and host certificates generated Ensure that the root and host certificates generated on the right box are uploaded correctly on the left box and vice versa.
- If the tunnel does not open, verify the parameters on both boxes. You can also check *Logs* > *System logs* > *IPSec logs* for further troubleshooting.
- Additional tunnels can be created to network multiple offices; the most important point to remember is to use different IP subnets for different offices.
- IKE aggressive mode (available under advanced options) is definitely not preferred; it transmits a pre-shared key in clear text.
- Though IPCop has an inbuilt certificate generator, you may also purchase and use certificates from various certifying authorities. END

References

- Website: *www.ipcop.org* and forum: *www.ipcops.com.*
- IPSec basics: *http://bit.ly/zXgoCN*
- NIST IPSec VPN guidelines: *http://1.usa.gov/efHo1b*

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